

Problem 1

Implement a feed-forward neural network in PyTorch.

1. Construct a synthetic regression dataset with a train and test dataset, where a one-layer network does not show good performance on the test dataset, while a network with more layers shows good performance.
2. Can you propose a regularization term that reduces the performance gap between the test and train datasets for the larger network? Show empirically that your regularizer works.

```
In [1]: import torch
import torch.nn as nn

class SyntheticRegressionDataset(torch.utils.data.Dataset):
    def __init__(self, num_samples, input_size):
        self.X = torch.randn(num_samples, input_size)
        # self.y = torch.sum(self.X, dim=1) * 0.4 + torch.randn(num_sampl
        self.y = torch.sin(torch.sum(self.X, dim=1)) + torch.randn(num_sa

    def __len__(self):
        return len(self.X)

    def __getitem__(self, idx):
        return self.X[idx], self.y[idx]
```

```
/home/vaishnavahari/26ss/(VU) Deep Learning for Structured Data/.venv/lib/
python3.13/site-packages/torch/_subclasses/functional_tensor.py:307: UserW
arning: Failed to initialize NumPy: No module named 'numpy' (Triggered int
ernally at /pytorch/torch/csrc/utils/tensor_numpy.cpp:84.)
  cpu = _conversion_method_template(device=torch.device("cpu"))
```

```
In [2]: class SimpleNN(nn.Module):
    def __init__(self, input_size, output_size):
        super(SimpleNN, self).__init__()
        self.fc1 = nn.Linear(input_size, output_size)

    def forward(self, x):
        out = self.fc1(x)
        return out

class MultiLayerNN(nn.Module):
    def __init__(self, input_size, hidden_size, output_size):
        super(MultiLayerNN, self).__init__()
        self.fc1 = nn.Linear(input_size, hidden_size)
        self.fc2 = nn.Linear(hidden_size, output_size)

    def forward(self, x):
        out = torch.relu(self.fc1(x))
        out = self.fc2(out)
        return out
```

```
In [3]: torch.manual_seed(0)
input_size = 10
```

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output_size = 1

dataset = SyntheticRegressionDataset(1000, input_size)
# test_dataset = SyntheticRegressionDataset(200, input_size)
# split dataset into train and test
train_dataset, test_dataset = torch.utils.data.random_split(dataset, [0.8

train_loader = torch.utils.data.DataLoader(train_dataset, batch_size=32,
test_loader = torch.utils.data.DataLoader(test_dataset, batch_size=32, sh

one_layer_model = SimpleNN(input_size, output_size)
criterion = nn.MSELoss()
optimizer = torch.optim.Adam(one_layer_model.parameters(), lr=0.01)

NUM_EPOCHS = 500
```

```
In [4]: # Train the one-layer model
for epoch in range(NUM_EPOCHS):
    train_loss = 0
    for X_batch, y_batch in train_loader:
        optimizer.zero_grad()
        outputs = one_layer_model(X_batch)
        loss = criterion(outputs.squeeze(), y_batch)
        loss.backward()
        optimizer.step()
        train_loss += loss.item() * X_batch.size(0)
    print(f'Epoch [{epoch+1}/{NUM_EPOCHS}], Loss: {train_loss / len(train_loader)}')
# Evaluate the one-layer model
one_layer_model.eval()
with torch.no_grad():
    test_loss = 0
    for X_batch, y_batch in test_loader:
        outputs = one_layer_model(X_batch)
        loss = criterion(outputs.squeeze(), y_batch)
        test_loss += loss.item() * X_batch.size(0)
print(f'One-layer model test loss: {test_loss / len(test_loader):.4f}')
one_layer_train_loss = train_loss / len(train_loader)
one_layer_test_loss = test_loss / len(test_loader)
```

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Epoch [463/500], Loss: 17.0247
Epoch [464/500], Loss: 17.0019
Epoch [465/500], Loss: 17.0681
Epoch [466/500], Loss: 16.9741
Epoch [467/500], Loss: 17.0308
Epoch [468/500], Loss: 17.0017
Epoch [469/500], Loss: 16.9433
Epoch [470/500], Loss: 16.9355
Epoch [471/500], Loss: 17.0629
Epoch [472/500], Loss: 17.0620
Epoch [473/500], Loss: 16.9780
Epoch [474/500], Loss: 17.0343
Epoch [475/500], Loss: 17.0645
Epoch [476/500], Loss: 16.9782
Epoch [477/500], Loss: 17.0115
Epoch [478/500], Loss: 16.9375
Epoch [479/500], Loss: 16.9652
Epoch [480/500], Loss: 16.9599
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Epoch [481/500], Loss: 17.0451
Epoch [482/500], Loss: 16.9468
Epoch [483/500], Loss: 16.9528
Epoch [484/500], Loss: 17.0527
Epoch [485/500], Loss: 16.9398
Epoch [486/500], Loss: 17.0280
Epoch [487/500], Loss: 16.9456
Epoch [488/500], Loss: 16.9174
Epoch [489/500], Loss: 16.9908
Epoch [490/500], Loss: 16.9844
Epoch [491/500], Loss: 16.9788
Epoch [492/500], Loss: 16.9726
Epoch [493/500], Loss: 16.9454
Epoch [494/500], Loss: 16.9892
Epoch [495/500], Loss: 16.9562
Epoch [496/500], Loss: 16.9378
Epoch [497/500], Loss: 17.0318
Epoch [498/500], Loss: 16.9620
Epoch [499/500], Loss: 16.9785
Epoch [500/500], Loss: 17.0110
One-layer model test loss: 15.6243

```

```

In [5]: # Train multi-layer model
        hidden_size = 20
        multi_layer_model = MultiLayerNN(input_size, hidden_size, output_size)
        optimizer = torch.optim.Adam(multi_layer_model.parameters(), lr=0.01)

```

```

In [6]: for epoch in range(NUM_EPOCHS):
        train_loss = 0
        for X_batch, y_batch in train_loader:
            optimizer.zero_grad()
            outputs = multi_layer_model(X_batch)
            loss = criterion(outputs.squeeze(), y_batch)
            loss.backward()
            optimizer.step()
            train_loss += loss.item() * X_batch.size(0)
        print(f'Epoch [{epoch+1}/{NUM_EPOCHS}], Loss: {train_loss / len(train_loader)}')
        # Evaluate the multi-layer model
        multi_layer_model.eval()
        with torch.no_grad():
            test_loss = 0
            for X_batch, y_batch in test_loader:
                outputs = multi_layer_model(X_batch)
                loss = criterion(outputs.squeeze(), y_batch)
                test_loss += loss.item() * X_batch.size(0)
        print(f'Multi-layer model test loss: {test_loss / len(test_loader):.4f}')
        multi_layer_train_loss = train_loss / len(train_loader)
        multi_layer_test_loss = test_loss / len(test_loader)

```

```
Epoch [1/500], Loss: 17.9806
Epoch [2/500], Loss: 17.0974
Epoch [3/500], Loss: 16.8079
Epoch [4/500], Loss: 16.5380
Epoch [5/500], Loss: 16.4141
Epoch [6/500], Loss: 16.4074
Epoch [7/500], Loss: 16.2053
Epoch [8/500], Loss: 16.2765
Epoch [9/500], Loss: 15.9806
Epoch [10/500], Loss: 16.0154
Epoch [11/500], Loss: 15.8053
Epoch [12/500], Loss: 15.5615
Epoch [13/500], Loss: 15.4202
Epoch [14/500], Loss: 15.5083
Epoch [15/500], Loss: 15.5534
Epoch [16/500], Loss: 15.3043
Epoch [17/500], Loss: 15.3519
Epoch [18/500], Loss: 14.8570
Epoch [19/500], Loss: 14.7034
Epoch [20/500], Loss: 14.0743
Epoch [21/500], Loss: 13.7380
Epoch [22/500], Loss: 13.2597
Epoch [23/500], Loss: 12.2158
Epoch [24/500], Loss: 11.4577
Epoch [25/500], Loss: 10.6664
Epoch [26/500], Loss: 10.3277
Epoch [27/500], Loss: 9.6524
Epoch [28/500], Loss: 9.7621
Epoch [29/500], Loss: 9.3816
Epoch [30/500], Loss: 9.4036
Epoch [31/500], Loss: 8.6973
Epoch [32/500], Loss: 8.4491
Epoch [33/500], Loss: 8.2772
Epoch [34/500], Loss: 8.7858
Epoch [35/500], Loss: 8.4185
Epoch [36/500], Loss: 8.1595
Epoch [37/500], Loss: 8.2596
Epoch [38/500], Loss: 7.9840
Epoch [39/500], Loss: 8.0775
Epoch [40/500], Loss: 8.0506
Epoch [41/500], Loss: 7.7743
Epoch [42/500], Loss: 7.7865
Epoch [43/500], Loss: 7.8884
Epoch [44/500], Loss: 7.2930
Epoch [45/500], Loss: 7.8808
Epoch [46/500], Loss: 7.5461
Epoch [47/500], Loss: 7.3532
Epoch [48/500], Loss: 7.3092
Epoch [49/500], Loss: 7.0677
Epoch [50/500], Loss: 7.4014
Epoch [51/500], Loss: 7.2889
Epoch [52/500], Loss: 6.9854
Epoch [53/500], Loss: 7.2735
Epoch [54/500], Loss: 7.2728
Epoch [55/500], Loss: 6.8721
Epoch [56/500], Loss: 7.2097
Epoch [57/500], Loss: 6.6785
Epoch [58/500], Loss: 6.8464
Epoch [59/500], Loss: 7.0025
Epoch [60/500], Loss: 6.8324
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Epoch [61/500], Loss: 6.2175
Epoch [62/500], Loss: 5.9817
Epoch [63/500], Loss: 5.6895
Epoch [64/500], Loss: 5.6474
Epoch [65/500], Loss: 5.9288
Epoch [66/500], Loss: 5.7290
Epoch [67/500], Loss: 5.3177
Epoch [68/500], Loss: 5.1455
Epoch [69/500], Loss: 5.1612
Epoch [70/500], Loss: 5.0546
Epoch [71/500], Loss: 4.7961
Epoch [72/500], Loss: 4.6097
Epoch [73/500], Loss: 4.6216
Epoch [74/500], Loss: 4.1970
Epoch [75/500], Loss: 4.0801
Epoch [76/500], Loss: 3.8577
Epoch [77/500], Loss: 3.8660
Epoch [78/500], Loss: 3.6669
Epoch [79/500], Loss: 3.6539
Epoch [80/500], Loss: 3.3645
Epoch [81/500], Loss: 3.4794
Epoch [82/500], Loss: 3.2461
Epoch [83/500], Loss: 3.0986
Epoch [84/500], Loss: 3.0427
Epoch [85/500], Loss: 2.7982
Epoch [86/500], Loss: 2.9056
Epoch [87/500], Loss: 2.9999
Epoch [88/500], Loss: 3.1333
Epoch [89/500], Loss: 2.8668
Epoch [90/500], Loss: 2.7732
Epoch [91/500], Loss: 2.9643
Epoch [92/500], Loss: 3.0051
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Epoch [94/500], Loss: 2.5440
Epoch [95/500], Loss: 2.3558
Epoch [96/500], Loss: 2.5431
Epoch [97/500], Loss: 2.2892
Epoch [98/500], Loss: 2.2686
Epoch [99/500], Loss: 2.3999
Epoch [100/500], Loss: 2.3806
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Epoch [102/500], Loss: 2.3561
Epoch [103/500], Loss: 2.1451
Epoch [104/500], Loss: 2.3413
Epoch [105/500], Loss: 2.5492
Epoch [106/500], Loss: 2.4518
Epoch [107/500], Loss: 2.4432
Epoch [108/500], Loss: 2.1017
Epoch [109/500], Loss: 2.3050
Epoch [110/500], Loss: 2.2208
Epoch [111/500], Loss: 2.2683
Epoch [112/500], Loss: 2.1502
Epoch [113/500], Loss: 2.0183
Epoch [114/500], Loss: 2.1199
Epoch [115/500], Loss: 2.2254
Epoch [116/500], Loss: 2.2710
Epoch [117/500], Loss: 2.0564
Epoch [118/500], Loss: 2.1259
Epoch [119/500], Loss: 2.0284
Epoch [120/500], Loss: 1.9399
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Epoch [126/500], Loss: 1.9570
Epoch [127/500], Loss: 1.8785
Epoch [128/500], Loss: 1.8976
Epoch [129/500], Loss: 1.8439
Epoch [130/500], Loss: 1.9621
Epoch [131/500], Loss: 1.9745
Epoch [132/500], Loss: 2.3711
Epoch [133/500], Loss: 1.9732
Epoch [134/500], Loss: 1.9354
Epoch [135/500], Loss: 1.9935
Epoch [136/500], Loss: 1.9968
Epoch [137/500], Loss: 2.3852
Epoch [138/500], Loss: 2.5482
Epoch [139/500], Loss: 2.0267
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Epoch [141/500], Loss: 2.0182
Epoch [142/500], Loss: 2.0508
Epoch [143/500], Loss: 2.0106
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Epoch [147/500], Loss: 1.8536
Epoch [148/500], Loss: 2.0348
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Epoch [150/500], Loss: 2.0400
Epoch [151/500], Loss: 2.0002
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Epoch [154/500], Loss: 1.9409
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Epoch [156/500], Loss: 1.9194
Epoch [157/500], Loss: 1.9592
Epoch [158/500], Loss: 1.9487
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Epoch [179/500], Loss: 1.6274
Epoch [180/500], Loss: 1.6572
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Epoch [240/500], Loss: 1.5405
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Epoch [241/500], Loss: 1.4994
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Epoch [249/500], Loss: 1.6679
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Epoch [257/500], Loss: 1.5300
Epoch [258/500], Loss: 1.6996
Epoch [259/500], Loss: 1.4463
Epoch [260/500], Loss: 1.5579
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Epoch [262/500], Loss: 1.5796
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Epoch [279/500], Loss: 1.5846
Epoch [280/500], Loss: 1.7481
Epoch [281/500], Loss: 1.5229
Epoch [282/500], Loss: 1.5148
Epoch [283/500], Loss: 1.4435
Epoch [284/500], Loss: 1.4466
Epoch [285/500], Loss: 1.6126
Epoch [286/500], Loss: 1.5755
Epoch [287/500], Loss: 1.5162
Epoch [288/500], Loss: 1.4961
Epoch [289/500], Loss: 1.4333
Epoch [290/500], Loss: 1.5173
Epoch [291/500], Loss: 1.4977
Epoch [292/500], Loss: 1.4373
Epoch [293/500], Loss: 1.4135
Epoch [294/500], Loss: 1.4927
Epoch [295/500], Loss: 1.7555
Epoch [296/500], Loss: 1.8689
Epoch [297/500], Loss: 1.6884
Epoch [298/500], Loss: 1.4684
Epoch [299/500], Loss: 1.3925
Epoch [300/500], Loss: 1.6095
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Epoch [301/500], Loss: 1.5230
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Epoch [303/500], Loss: 1.7068
Epoch [304/500], Loss: 1.4959
Epoch [305/500], Loss: 1.5912
Epoch [306/500], Loss: 1.4970
Epoch [307/500], Loss: 1.5397
Epoch [308/500], Loss: 1.5541
Epoch [309/500], Loss: 1.5856
Epoch [310/500], Loss: 1.5651
Epoch [311/500], Loss: 1.5535
Epoch [312/500], Loss: 1.6068
Epoch [313/500], Loss: 1.5206
Epoch [314/500], Loss: 1.4052
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Epoch [320/500], Loss: 1.5083
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Epoch [322/500], Loss: 1.5600
Epoch [323/500], Loss: 1.4372
Epoch [324/500], Loss: 1.4681
Epoch [325/500], Loss: 1.5492
Epoch [326/500], Loss: 1.3891
Epoch [327/500], Loss: 1.5173
Epoch [328/500], Loss: 1.3329
Epoch [329/500], Loss: 1.4155
Epoch [330/500], Loss: 1.4163
Epoch [331/500], Loss: 1.3834
Epoch [332/500], Loss: 1.4463
Epoch [333/500], Loss: 1.3927
Epoch [334/500], Loss: 1.3581
Epoch [335/500], Loss: 1.4002
Epoch [336/500], Loss: 1.4683
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Epoch [338/500], Loss: 1.7440
Epoch [339/500], Loss: 1.6119
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Epoch [343/500], Loss: 1.4991
Epoch [344/500], Loss: 1.3957
Epoch [345/500], Loss: 1.3743
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Epoch [351/500], Loss: 1.5203
Epoch [352/500], Loss: 1.5386
Epoch [353/500], Loss: 1.7704
Epoch [354/500], Loss: 1.9663
Epoch [355/500], Loss: 1.5277
Epoch [356/500], Loss: 1.5799
Epoch [357/500], Loss: 1.6317
Epoch [358/500], Loss: 1.4276
Epoch [359/500], Loss: 1.4511
Epoch [360/500], Loss: 1.4745
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Epoch [361/500], Loss: 1.4767
Epoch [362/500], Loss: 1.3801
Epoch [363/500], Loss: 1.5480
Epoch [364/500], Loss: 1.7005
Epoch [365/500], Loss: 1.5660
Epoch [366/500], Loss: 1.3359
Epoch [367/500], Loss: 1.4354
Epoch [368/500], Loss: 1.4469
Epoch [369/500], Loss: 1.4031
Epoch [370/500], Loss: 1.5843
Epoch [371/500], Loss: 1.6115
Epoch [372/500], Loss: 1.6769
Epoch [373/500], Loss: 1.5145
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Epoch [375/500], Loss: 1.5704
Epoch [376/500], Loss: 1.4997
Epoch [377/500], Loss: 1.6424
Epoch [378/500], Loss: 1.6735
Epoch [379/500], Loss: 1.7692
Epoch [380/500], Loss: 1.6162
Epoch [381/500], Loss: 1.5674
Epoch [382/500], Loss: 1.8227
Epoch [383/500], Loss: 1.4708
Epoch [384/500], Loss: 1.4273
Epoch [385/500], Loss: 1.4088
Epoch [386/500], Loss: 1.6848
Epoch [387/500], Loss: 1.5603
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Epoch [389/500], Loss: 1.3723
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Epoch [391/500], Loss: 1.4818
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Epoch [393/500], Loss: 1.8239
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Epoch [406/500], Loss: 1.6398
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Epoch [409/500], Loss: 1.5003
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Epoch [411/500], Loss: 1.4325
Epoch [412/500], Loss: 1.3117
Epoch [413/500], Loss: 1.2862
Epoch [414/500], Loss: 1.2607
Epoch [415/500], Loss: 1.2742
Epoch [416/500], Loss: 1.3121
Epoch [417/500], Loss: 1.4053
Epoch [418/500], Loss: 1.2303
Epoch [419/500], Loss: 1.2994
Epoch [420/500], Loss: 1.3432
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Epoch [422/500], Loss: 1.2720
Epoch [423/500], Loss: 1.2999
Epoch [424/500], Loss: 1.3248
Epoch [425/500], Loss: 1.4108
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Epoch [427/500], Loss: 1.6020
Epoch [428/500], Loss: 1.4882
Epoch [429/500], Loss: 1.4106
Epoch [430/500], Loss: 1.8176
Epoch [431/500], Loss: 1.6566
Epoch [432/500], Loss: 1.6712
Epoch [433/500], Loss: 1.5865
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Epoch [435/500], Loss: 1.4931
Epoch [436/500], Loss: 1.5672
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Epoch [452/500], Loss: 1.3280
Epoch [453/500], Loss: 1.3317
Epoch [454/500], Loss: 1.3779
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Epoch [456/500], Loss: 1.4539
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Epoch [458/500], Loss: 1.4399
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Epoch [460/500], Loss: 1.3385
Epoch [461/500], Loss: 1.6148
Epoch [462/500], Loss: 1.5625
Epoch [463/500], Loss: 1.4544
Epoch [464/500], Loss: 1.3572
Epoch [465/500], Loss: 1.3360
Epoch [466/500], Loss: 1.3682
Epoch [467/500], Loss: 1.5490
Epoch [468/500], Loss: 1.4911
Epoch [469/500], Loss: 1.3730
Epoch [470/500], Loss: 1.3159
Epoch [471/500], Loss: 1.4793
Epoch [472/500], Loss: 1.4596
Epoch [473/500], Loss: 1.2904
Epoch [474/500], Loss: 1.2685
Epoch [475/500], Loss: 1.2618
Epoch [476/500], Loss: 1.3430
Epoch [477/500], Loss: 1.4751
Epoch [478/500], Loss: 1.5981
Epoch [479/500], Loss: 1.5956
Epoch [480/500], Loss: 1.5367
```

```
Epoch [481/500], Loss: 1.3555
Epoch [482/500], Loss: 1.4336
Epoch [483/500], Loss: 1.2799
Epoch [484/500], Loss: 1.3628
Epoch [485/500], Loss: 1.4227
Epoch [486/500], Loss: 1.2934
Epoch [487/500], Loss: 1.3807
Epoch [488/500], Loss: 1.2919
Epoch [489/500], Loss: 1.3785
Epoch [490/500], Loss: 1.3226
Epoch [491/500], Loss: 1.3032
Epoch [492/500], Loss: 1.3042
Epoch [493/500], Loss: 1.6454
Epoch [494/500], Loss: 1.2929
Epoch [495/500], Loss: 1.3495
Epoch [496/500], Loss: 1.4668
Epoch [497/500], Loss: 1.5846
Epoch [498/500], Loss: 1.3172
Epoch [499/500], Loss: 1.3724
Epoch [500/500], Loss: 1.3003
Multi-layer model test loss: 4.7986
```

```
In [7]: print(f'One-layer model train loss: {one_layer_train_loss:.4f}, test loss: {one_layer_test_loss:.4f}')
        print(f'Multi-layer model train loss: {multi_layer_train_loss:.4f}, test loss: {multi_layer_test_loss:.4f}')
```

```
One-layer model train loss: 17.0110, test loss: 15.6243
Multi-layer model train loss: 1.3003, test loss: 4.7986
```

Now with L2 regularization:

```
In [8]: # Regularization L2
        l2_lambda = 0.0001
        multi_layer_model = MultiLayerNN(input_size, hidden_size, output_size)
        optimizer = torch.optim.Adam(multi_layer_model.parameters(), lr=0.01, weight_decay=l2_lambda)
```

```
In [9]: for epoch in range(NUM_EPOCHS):
        train_loss = 0
        for X_batch, y_batch in train_loader:
            optimizer.zero_grad()
            outputs = multi_layer_model(X_batch)
            loss = criterion(outputs.squeeze(), y_batch)
            loss.backward()
            optimizer.step()
            train_loss += loss.item() * X_batch.size(0)
        print(f'Epoch [{epoch+1}/{NUM_EPOCHS}], Loss: {train_loss / len(train_loader):.4f}')
        multi_layer_model.eval()
        with torch.no_grad():
            test_loss = 0
            for X_batch, y_batch in test_loader:
                outputs = multi_layer_model(X_batch)
                loss = criterion(outputs.squeeze(), y_batch)
                test_loss += loss.item() * X_batch.size(0)
            print(f'Multi-layer model test loss: {test_loss / len(test_loader):.4f}')
        multi_layer_regularized_train_loss = train_loss / len(train_loader)
        multi_layer_regularized_test_loss = test_loss / len(test_loader)
```

```
Epoch [1/500], Loss: 17.7210
Epoch [2/500], Loss: 17.0754
Epoch [3/500], Loss: 16.7207
Epoch [4/500], Loss: 16.5178
Epoch [5/500], Loss: 16.3680
Epoch [6/500], Loss: 16.2418
Epoch [7/500], Loss: 16.4118
Epoch [8/500], Loss: 15.9732
Epoch [9/500], Loss: 16.0440
Epoch [10/500], Loss: 15.8614
Epoch [11/500], Loss: 15.7509
Epoch [12/500], Loss: 15.7333
Epoch [13/500], Loss: 15.5892
Epoch [14/500], Loss: 15.7841
Epoch [15/500], Loss: 15.7157
Epoch [16/500], Loss: 15.2041
Epoch [17/500], Loss: 15.3495
Epoch [18/500], Loss: 15.3995
Epoch [19/500], Loss: 15.0712
Epoch [20/500], Loss: 15.1286
Epoch [21/500], Loss: 15.1597
Epoch [22/500], Loss: 14.8997
Epoch [23/500], Loss: 14.9521
Epoch [24/500], Loss: 14.9389
Epoch [25/500], Loss: 14.8881
Epoch [26/500], Loss: 14.9065
Epoch [27/500], Loss: 14.7707
Epoch [28/500], Loss: 14.7657
Epoch [29/500], Loss: 14.4128
Epoch [30/500], Loss: 14.3590
Epoch [31/500], Loss: 14.4928
Epoch [32/500], Loss: 14.6321
Epoch [33/500], Loss: 14.4581
Epoch [34/500], Loss: 14.5835
Epoch [35/500], Loss: 14.0170
Epoch [36/500], Loss: 14.6097
Epoch [37/500], Loss: 14.0710
Epoch [38/500], Loss: 13.8818
Epoch [39/500], Loss: 14.0111
Epoch [40/500], Loss: 13.9640
Epoch [41/500], Loss: 13.7394
Epoch [42/500], Loss: 13.9298
Epoch [43/500], Loss: 13.6360
Epoch [44/500], Loss: 13.7330
Epoch [45/500], Loss: 13.2125
Epoch [46/500], Loss: 13.3449
Epoch [47/500], Loss: 13.3560
Epoch [48/500], Loss: 13.0760
Epoch [49/500], Loss: 13.0985
Epoch [50/500], Loss: 12.8353
Epoch [51/500], Loss: 12.6881
Epoch [52/500], Loss: 12.6109
Epoch [53/500], Loss: 12.9058
Epoch [54/500], Loss: 12.2861
Epoch [55/500], Loss: 12.6264
Epoch [56/500], Loss: 12.8599
Epoch [57/500], Loss: 12.3554
Epoch [58/500], Loss: 12.2296
Epoch [59/500], Loss: 12.0929
Epoch [60/500], Loss: 11.8188
```

```
Epoch [61/500], Loss: 11.8477
Epoch [62/500], Loss: 11.4160
Epoch [63/500], Loss: 10.9154
Epoch [64/500], Loss: 10.3915
Epoch [65/500], Loss: 9.7767
Epoch [66/500], Loss: 9.3955
Epoch [67/500], Loss: 8.9179
Epoch [68/500], Loss: 8.8213
Epoch [69/500], Loss: 8.6039
Epoch [70/500], Loss: 8.2566
Epoch [71/500], Loss: 8.0851
Epoch [72/500], Loss: 7.9993
Epoch [73/500], Loss: 7.6112
Epoch [74/500], Loss: 7.7388
Epoch [75/500], Loss: 8.3380
Epoch [76/500], Loss: 7.5076
Epoch [77/500], Loss: 7.7399
Epoch [78/500], Loss: 7.7947
Epoch [79/500], Loss: 7.3265
Epoch [80/500], Loss: 7.4806
Epoch [81/500], Loss: 7.6756
Epoch [82/500], Loss: 7.2704
Epoch [83/500], Loss: 7.0830
Epoch [84/500], Loss: 7.4680
Epoch [85/500], Loss: 7.0983
Epoch [86/500], Loss: 7.3640
Epoch [87/500], Loss: 7.0322
Epoch [88/500], Loss: 7.4180
Epoch [89/500], Loss: 6.9037
Epoch [90/500], Loss: 7.3010
Epoch [91/500], Loss: 7.4126
Epoch [92/500], Loss: 7.4162
Epoch [93/500], Loss: 7.1608
Epoch [94/500], Loss: 6.7020
Epoch [95/500], Loss: 7.0588
Epoch [96/500], Loss: 6.8237
Epoch [97/500], Loss: 6.8450
Epoch [98/500], Loss: 7.1723
Epoch [99/500], Loss: 6.9513
Epoch [100/500], Loss: 6.6153
Epoch [101/500], Loss: 6.8355
Epoch [102/500], Loss: 7.1756
Epoch [103/500], Loss: 6.5416
Epoch [104/500], Loss: 6.7176
Epoch [105/500], Loss: 6.8339
Epoch [106/500], Loss: 6.3271
Epoch [107/500], Loss: 6.1572
Epoch [108/500], Loss: 6.1688
Epoch [109/500], Loss: 5.9551
Epoch [110/500], Loss: 5.8188
Epoch [111/500], Loss: 6.1187
Epoch [112/500], Loss: 5.6561
Epoch [113/500], Loss: 5.4660
Epoch [114/500], Loss: 5.5200
Epoch [115/500], Loss: 5.9347
Epoch [116/500], Loss: 5.5153
Epoch [117/500], Loss: 5.1386
Epoch [118/500], Loss: 5.0951
Epoch [119/500], Loss: 5.0048
Epoch [120/500], Loss: 5.2007
```

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Epoch [121/500], Loss: 5.1142
Epoch [122/500], Loss: 5.1866
Epoch [123/500], Loss: 5.1653
Epoch [124/500], Loss: 5.1036
Epoch [125/500], Loss: 5.1864
Epoch [126/500], Loss: 4.6708
Epoch [127/500], Loss: 4.6035
Epoch [128/500], Loss: 4.7142
Epoch [129/500], Loss: 4.7767
Epoch [130/500], Loss: 4.5397
Epoch [131/500], Loss: 4.4196
Epoch [132/500], Loss: 4.3679
Epoch [133/500], Loss: 4.4781
Epoch [134/500], Loss: 4.4958
Epoch [135/500], Loss: 4.3675
Epoch [136/500], Loss: 4.3698
Epoch [137/500], Loss: 4.4523
Epoch [138/500], Loss: 4.3042
Epoch [139/500], Loss: 4.3437
Epoch [140/500], Loss: 4.2655
Epoch [141/500], Loss: 4.3737
Epoch [142/500], Loss: 4.4135
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Epoch [144/500], Loss: 4.3581
Epoch [145/500], Loss: 4.2304
Epoch [146/500], Loss: 4.1906
Epoch [147/500], Loss: 4.1364
Epoch [148/500], Loss: 4.2377
Epoch [149/500], Loss: 4.3742
Epoch [150/500], Loss: 4.1296
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Epoch [152/500], Loss: 4.0703
Epoch [153/500], Loss: 4.3019
Epoch [154/500], Loss: 4.0942
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Epoch [156/500], Loss: 4.1920
Epoch [157/500], Loss: 4.1150
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Epoch [163/500], Loss: 4.1749
Epoch [164/500], Loss: 3.9964
Epoch [165/500], Loss: 4.0712
Epoch [166/500], Loss: 4.1469
Epoch [167/500], Loss: 4.1741
Epoch [168/500], Loss: 4.3703
Epoch [169/500], Loss: 4.1172
Epoch [170/500], Loss: 4.1888
Epoch [171/500], Loss: 3.9974
Epoch [172/500], Loss: 3.9582
Epoch [173/500], Loss: 3.9872
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Epoch [175/500], Loss: 4.2201
Epoch [176/500], Loss: 4.4244
Epoch [177/500], Loss: 4.1434
Epoch [178/500], Loss: 3.7000
Epoch [179/500], Loss: 3.5785
Epoch [180/500], Loss: 3.6811
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Epoch [184/500], Loss: 3.7061
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Epoch [191/500], Loss: 3.4412
Epoch [192/500], Loss: 3.5018
Epoch [193/500], Loss: 3.3189
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Epoch [201/500], Loss: 3.2972
Epoch [202/500], Loss: 3.2570
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Epoch [205/500], Loss: 2.9303
Epoch [206/500], Loss: 2.9567
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Epoch [208/500], Loss: 2.8180
Epoch [209/500], Loss: 2.8603
Epoch [210/500], Loss: 2.8307
Epoch [211/500], Loss: 2.8513
Epoch [212/500], Loss: 2.7396
Epoch [213/500], Loss: 2.7297
Epoch [214/500], Loss: 2.7486
Epoch [215/500], Loss: 3.0038
Epoch [216/500], Loss: 2.9724
Epoch [217/500], Loss: 2.5681
Epoch [218/500], Loss: 2.6048
Epoch [219/500], Loss: 2.6670
Epoch [220/500], Loss: 2.6030
Epoch [221/500], Loss: 2.5624
Epoch [222/500], Loss: 2.6111
Epoch [223/500], Loss: 2.6882
Epoch [224/500], Loss: 2.5618
Epoch [225/500], Loss: 2.8619
Epoch [226/500], Loss: 2.6966
Epoch [227/500], Loss: 2.5181
Epoch [228/500], Loss: 2.5299
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Epoch [230/500], Loss: 2.5802
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Epoch [233/500], Loss: 2.4562
Epoch [234/500], Loss: 2.5424
Epoch [235/500], Loss: 2.5584
Epoch [236/500], Loss: 2.7702
Epoch [237/500], Loss: 2.8675
Epoch [238/500], Loss: 2.5208
Epoch [239/500], Loss: 2.5124
Epoch [240/500], Loss: 2.4841
```



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Epoch [241/500], Loss: 2.5519
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Epoch [243/500], Loss: 2.4555
Epoch [244/500], Loss: 2.7019
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Epoch [246/500], Loss: 2.7763
Epoch [247/500], Loss: 2.6026
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Epoch [250/500], Loss: 2.4416
Epoch [251/500], Loss: 2.4925
Epoch [252/500], Loss: 2.3874
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Epoch [279/500], Loss: 2.3830
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Epoch [283/500], Loss: 2.1966
Epoch [284/500], Loss: 2.1921
Epoch [285/500], Loss: 2.2341
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Epoch [287/500], Loss: 2.1852
Epoch [288/500], Loss: 2.4588
Epoch [289/500], Loss: 2.3502
Epoch [290/500], Loss: 2.2740
Epoch [291/500], Loss: 2.3725
Epoch [292/500], Loss: 2.3012
Epoch [293/500], Loss: 2.2039
Epoch [294/500], Loss: 2.5559
Epoch [295/500], Loss: 2.5793
Epoch [296/500], Loss: 2.3491
Epoch [297/500], Loss: 2.3840
Epoch [298/500], Loss: 2.3675
Epoch [299/500], Loss: 2.1897
Epoch [300/500], Loss: 2.1709
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Epoch [301/500], Loss: 2.2231
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Epoch [304/500], Loss: 2.3214
Epoch [305/500], Loss: 2.1557
Epoch [306/500], Loss: 2.2218
Epoch [307/500], Loss: 2.2986
Epoch [308/500], Loss: 2.4151
Epoch [309/500], Loss: 2.2383
Epoch [310/500], Loss: 2.3781
Epoch [311/500], Loss: 2.2696
Epoch [312/500], Loss: 2.2935
Epoch [313/500], Loss: 2.2931
Epoch [314/500], Loss: 3.0138
Epoch [315/500], Loss: 2.3594
Epoch [316/500], Loss: 2.1164
Epoch [317/500], Loss: 2.0909
Epoch [318/500], Loss: 2.0880
Epoch [319/500], Loss: 2.3641
Epoch [320/500], Loss: 2.5886
Epoch [321/500], Loss: 2.1614
Epoch [322/500], Loss: 2.2154
Epoch [323/500], Loss: 2.1895
Epoch [324/500], Loss: 2.1625
Epoch [325/500], Loss: 2.1880
Epoch [326/500], Loss: 2.3709
Epoch [327/500], Loss: 2.3411
Epoch [328/500], Loss: 2.4180
Epoch [329/500], Loss: 2.6366
Epoch [330/500], Loss: 2.1430
Epoch [331/500], Loss: 2.2273
Epoch [332/500], Loss: 2.1392
Epoch [333/500], Loss: 2.3480
Epoch [334/500], Loss: 2.3154
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Epoch [336/500], Loss: 2.2819
Epoch [337/500], Loss: 2.3736
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Epoch [342/500], Loss: 2.2019
Epoch [343/500], Loss: 2.3872
Epoch [344/500], Loss: 2.5176
Epoch [345/500], Loss: 2.5398
Epoch [346/500], Loss: 2.1267
Epoch [347/500], Loss: 2.1700
Epoch [348/500], Loss: 2.2385
Epoch [349/500], Loss: 2.2964
Epoch [350/500], Loss: 2.6006
Epoch [351/500], Loss: 2.6284
Epoch [352/500], Loss: 2.3801
Epoch [353/500], Loss: 2.4165
Epoch [354/500], Loss: 2.2755
Epoch [355/500], Loss: 2.2094
Epoch [356/500], Loss: 2.2348
Epoch [357/500], Loss: 2.1381
Epoch [358/500], Loss: 2.1715
Epoch [359/500], Loss: 2.2967
Epoch [360/500], Loss: 2.4399
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Epoch [364/500], Loss: 2.3699
Epoch [365/500], Loss: 2.3699
Epoch [366/500], Loss: 2.3040
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Epoch [368/500], Loss: 2.1676
Epoch [369/500], Loss: 2.1268
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Epoch [371/500], Loss: 2.1750
Epoch [372/500], Loss: 2.3392
Epoch [373/500], Loss: 2.2295
Epoch [374/500], Loss: 2.0534
Epoch [375/500], Loss: 2.1431
Epoch [376/500], Loss: 2.4179
Epoch [377/500], Loss: 2.2722
Epoch [378/500], Loss: 2.2654
Epoch [379/500], Loss: 2.2603
Epoch [380/500], Loss: 2.2449
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Epoch [383/500], Loss: 2.1091
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Epoch [385/500], Loss: 2.2728
Epoch [386/500], Loss: 2.2757
Epoch [387/500], Loss: 2.0577
Epoch [388/500], Loss: 2.0054
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Epoch [392/500], Loss: 2.1301
Epoch [393/500], Loss: 2.1645
Epoch [394/500], Loss: 2.4958
Epoch [395/500], Loss: 2.1117
Epoch [396/500], Loss: 2.0816
Epoch [397/500], Loss: 2.0187
Epoch [398/500], Loss: 2.0270
Epoch [399/500], Loss: 2.0197
Epoch [400/500], Loss: 2.2672
Epoch [401/500], Loss: 2.3538
Epoch [402/500], Loss: 2.2475
Epoch [403/500], Loss: 2.2449
Epoch [404/500], Loss: 2.2766
Epoch [405/500], Loss: 2.3572
Epoch [406/500], Loss: 2.1667
Epoch [407/500], Loss: 2.0486
Epoch [408/500], Loss: 2.1964
Epoch [409/500], Loss: 2.1477
Epoch [410/500], Loss: 2.1162
Epoch [411/500], Loss: 2.0131
Epoch [412/500], Loss: 2.1902
Epoch [413/500], Loss: 2.3678
Epoch [414/500], Loss: 2.4682
Epoch [415/500], Loss: 2.1673
Epoch [416/500], Loss: 2.0335
Epoch [417/500], Loss: 2.2850
Epoch [418/500], Loss: 2.4992
Epoch [419/500], Loss: 2.3280
Epoch [420/500], Loss: 2.0937
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Epoch [423/500], Loss: 2.1361
Epoch [424/500], Loss: 2.1407
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Epoch [427/500], Loss: 1.9608
Epoch [428/500], Loss: 1.9953
Epoch [429/500], Loss: 2.3098
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Epoch [431/500], Loss: 2.2138
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Epoch [437/500], Loss: 2.1059
Epoch [438/500], Loss: 2.1626
Epoch [439/500], Loss: 2.1388
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Epoch [444/500], Loss: 1.9215
Epoch [445/500], Loss: 1.9753
Epoch [446/500], Loss: 1.9473
Epoch [447/500], Loss: 1.9752
Epoch [448/500], Loss: 1.9373
Epoch [449/500], Loss: 1.9831
Epoch [450/500], Loss: 1.9200
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Epoch [452/500], Loss: 2.0355
Epoch [453/500], Loss: 2.0223
Epoch [454/500], Loss: 2.0866
Epoch [455/500], Loss: 1.9961
Epoch [456/500], Loss: 1.8950
Epoch [457/500], Loss: 1.9804
Epoch [458/500], Loss: 2.1334
Epoch [459/500], Loss: 2.0608
Epoch [460/500], Loss: 1.9900
Epoch [461/500], Loss: 1.9266
Epoch [462/500], Loss: 1.9734
Epoch [463/500], Loss: 1.8731
Epoch [464/500], Loss: 2.0751
Epoch [465/500], Loss: 1.9909
Epoch [466/500], Loss: 1.8918
Epoch [467/500], Loss: 1.9448
Epoch [468/500], Loss: 2.0748
Epoch [469/500], Loss: 1.9725
Epoch [470/500], Loss: 2.0039
Epoch [471/500], Loss: 2.1408
Epoch [472/500], Loss: 2.0868
Epoch [473/500], Loss: 1.8785
Epoch [474/500], Loss: 1.9398
Epoch [475/500], Loss: 1.9800
Epoch [476/500], Loss: 1.9221
Epoch [477/500], Loss: 1.8832
Epoch [478/500], Loss: 1.8287
Epoch [479/500], Loss: 1.8515
Epoch [480/500], Loss: 1.9659
```

```

Epoch [481/500], Loss: 2.0276
Epoch [482/500], Loss: 1.8207
Epoch [483/500], Loss: 1.8968
Epoch [484/500], Loss: 1.7570
Epoch [485/500], Loss: 1.9518
Epoch [486/500], Loss: 2.1755
Epoch [487/500], Loss: 2.1216
Epoch [488/500], Loss: 1.7779
Epoch [489/500], Loss: 2.0193
Epoch [490/500], Loss: 1.9768
Epoch [491/500], Loss: 1.9679
Epoch [492/500], Loss: 1.9613
Epoch [493/500], Loss: 1.7877
Epoch [494/500], Loss: 1.8382
Epoch [495/500], Loss: 1.7120
Epoch [496/500], Loss: 1.8335
Epoch [497/500], Loss: 2.4734
Epoch [498/500], Loss: 2.1099
Epoch [499/500], Loss: 2.3284
Epoch [500/500], Loss: 1.9108
Multi-layer model test loss: 4.4801

```

```

In [10]: print(f'Multi-layer model with L2 regularization train loss: {multi_layer
print(f'Multi-layer model without regularization train loss: {multi_layer

```

```

Multi-layer model with L2 regularization train loss: 1.9108, test loss:
4.4801
Multi-layer model without regularization train loss: 1.3003, test loss:
4.7986

```

Test loss is lower with regularization, even with the penalty on the weights. If trained further, the difference is more visible.

```

In [11]: for epoch in range(450):
        train_loss = 0
        for X_batch, y_batch in train_loader:
            optimizer.zero_grad()
            outputs = multi_layer_model(X_batch)
            loss = criterion(outputs.squeeze(), y_batch)
            loss.backward()
            optimizer.step()
            train_loss += loss.item() * X_batch.size(0)
        print(f'Epoch [{epoch+1}/450], Loss: {train_loss / len(train_loader):
multi_layer_model.eval()
        with torch.no_grad():
            test_loss = 0
            for X_batch, y_batch in test_loader:
                outputs = multi_layer_model(X_batch)
                loss = criterion(outputs.squeeze(), y_batch)
                test_loss += loss.item() * X_batch.size(0)
        print(f'Multi-layer model test loss: {test_loss / len(test_loader):.4f}')
        multi_layer_regularized_train_loss = train_loss / len(train_loader)
        multi_layer_regularized_test_loss = test_loss / len(test_loader)

```

```
Epoch [1/450], Loss: 2.0182
Epoch [2/450], Loss: 1.8769
Epoch [3/450], Loss: 1.8658
Epoch [4/450], Loss: 1.8648
Epoch [5/450], Loss: 1.9586
Epoch [6/450], Loss: 2.0106
Epoch [7/450], Loss: 2.0493
Epoch [8/450], Loss: 1.9525
Epoch [9/450], Loss: 1.8969
Epoch [10/450], Loss: 2.0597
Epoch [11/450], Loss: 1.8863
Epoch [12/450], Loss: 1.8091
Epoch [13/450], Loss: 1.9961
Epoch [14/450], Loss: 2.1160
Epoch [15/450], Loss: 1.8458
Epoch [16/450], Loss: 1.7602
Epoch [17/450], Loss: 1.8933
Epoch [18/450], Loss: 1.6909
Epoch [19/450], Loss: 1.7687
Epoch [20/450], Loss: 1.6807
Epoch [21/450], Loss: 1.7673
Epoch [22/450], Loss: 1.8823
Epoch [23/450], Loss: 1.8637
Epoch [24/450], Loss: 1.8344
Epoch [25/450], Loss: 1.8561
Epoch [26/450], Loss: 1.7369
Epoch [27/450], Loss: 1.7483
Epoch [28/450], Loss: 1.7503
Epoch [29/450], Loss: 1.8273
Epoch [30/450], Loss: 1.7659
Epoch [31/450], Loss: 1.7696
Epoch [32/450], Loss: 1.7996
Epoch [33/450], Loss: 1.7653
Epoch [34/450], Loss: 1.8452
Epoch [35/450], Loss: 1.7115
Epoch [36/450], Loss: 1.7704
Epoch [37/450], Loss: 1.9037
Epoch [38/450], Loss: 1.7756
Epoch [39/450], Loss: 1.7276
Epoch [40/450], Loss: 1.8969
Epoch [41/450], Loss: 1.7935
Epoch [42/450], Loss: 1.6785
Epoch [43/450], Loss: 1.8773
Epoch [44/450], Loss: 1.8376
Epoch [45/450], Loss: 1.8838
Epoch [46/450], Loss: 1.7704
Epoch [47/450], Loss: 1.6558
Epoch [48/450], Loss: 1.8504
Epoch [49/450], Loss: 1.8663
Epoch [50/450], Loss: 1.8013
Epoch [51/450], Loss: 1.8519
Epoch [52/450], Loss: 1.8350
Epoch [53/450], Loss: 1.9031
Epoch [54/450], Loss: 1.9846
Epoch [55/450], Loss: 1.9693
Epoch [56/450], Loss: 1.7932
Epoch [57/450], Loss: 1.7429
Epoch [58/450], Loss: 1.9092
Epoch [59/450], Loss: 1.8232
Epoch [60/450], Loss: 1.7929
```

```
Epoch [61/450], Loss: 2.1067
Epoch [62/450], Loss: 1.9074
Epoch [63/450], Loss: 1.9490
Epoch [64/450], Loss: 1.8435
Epoch [65/450], Loss: 1.7786
Epoch [66/450], Loss: 1.7874
Epoch [67/450], Loss: 1.9378
Epoch [68/450], Loss: 2.2509
Epoch [69/450], Loss: 1.8187
Epoch [70/450], Loss: 1.7487
Epoch [71/450], Loss: 1.8691
Epoch [72/450], Loss: 1.7801
Epoch [73/450], Loss: 1.8526
Epoch [74/450], Loss: 1.6942
Epoch [75/450], Loss: 1.7319
Epoch [76/450], Loss: 1.7442
Epoch [77/450], Loss: 1.7670
Epoch [78/450], Loss: 1.7523
Epoch [79/450], Loss: 1.8481
Epoch [80/450], Loss: 1.6590
Epoch [81/450], Loss: 1.8283
Epoch [82/450], Loss: 1.7493
Epoch [83/450], Loss: 1.8200
Epoch [84/450], Loss: 1.8983
Epoch [85/450], Loss: 1.8569
Epoch [86/450], Loss: 1.8247
Epoch [87/450], Loss: 1.8938
Epoch [88/450], Loss: 2.1020
Epoch [89/450], Loss: 2.2855
Epoch [90/450], Loss: 2.1069
Epoch [91/450], Loss: 1.9152
Epoch [92/450], Loss: 1.8783
Epoch [93/450], Loss: 1.7225
Epoch [94/450], Loss: 1.7108
Epoch [95/450], Loss: 1.7478
Epoch [96/450], Loss: 1.9053
Epoch [97/450], Loss: 1.7409
Epoch [98/450], Loss: 1.6729
Epoch [99/450], Loss: 1.7825
Epoch [100/450], Loss: 1.6678
Epoch [101/450], Loss: 1.7552
Epoch [102/450], Loss: 1.8981
Epoch [103/450], Loss: 1.7185
Epoch [104/450], Loss: 1.7417
Epoch [105/450], Loss: 1.6548
Epoch [106/450], Loss: 1.7689
Epoch [107/450], Loss: 1.7809
Epoch [108/450], Loss: 1.9209
Epoch [109/450], Loss: 1.7224
Epoch [110/450], Loss: 1.6191
Epoch [111/450], Loss: 1.6500
Epoch [112/450], Loss: 1.7444
Epoch [113/450], Loss: 1.8726
Epoch [114/450], Loss: 1.7632
Epoch [115/450], Loss: 1.8121
Epoch [116/450], Loss: 1.9302
Epoch [117/450], Loss: 2.2289
Epoch [118/450], Loss: 2.0153
Epoch [119/450], Loss: 1.8805
Epoch [120/450], Loss: 1.6711
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```
Epoch [121/450], Loss: 1.6119
Epoch [122/450], Loss: 1.7565
Epoch [123/450], Loss: 1.7952
Epoch [124/450], Loss: 1.8667
Epoch [125/450], Loss: 1.8171
Epoch [126/450], Loss: 1.7755
Epoch [127/450], Loss: 1.6819
Epoch [128/450], Loss: 1.9827
Epoch [129/450], Loss: 1.8874
Epoch [130/450], Loss: 1.8569
Epoch [131/450], Loss: 1.7984
Epoch [132/450], Loss: 1.8181
Epoch [133/450], Loss: 1.7317
Epoch [134/450], Loss: 2.0293
Epoch [135/450], Loss: 2.2245
Epoch [136/450], Loss: 1.9342
Epoch [137/450], Loss: 1.8506
Epoch [138/450], Loss: 1.6847
Epoch [139/450], Loss: 1.6430
Epoch [140/450], Loss: 1.6301
Epoch [141/450], Loss: 1.7310
Epoch [142/450], Loss: 2.3261
Epoch [143/450], Loss: 2.0552
Epoch [144/450], Loss: 2.0967
Epoch [145/450], Loss: 2.0792
Epoch [146/450], Loss: 1.5905
Epoch [147/450], Loss: 1.6150
Epoch [148/450], Loss: 1.6548
Epoch [149/450], Loss: 1.6682
Epoch [150/450], Loss: 1.7214
Epoch [151/450], Loss: 1.6728
Epoch [152/450], Loss: 1.7370
Epoch [153/450], Loss: 1.9582
Epoch [154/450], Loss: 2.0308
Epoch [155/450], Loss: 1.7693
Epoch [156/450], Loss: 1.8411
Epoch [157/450], Loss: 1.7639
Epoch [158/450], Loss: 1.6281
Epoch [159/450], Loss: 1.6542
Epoch [160/450], Loss: 1.7917
Epoch [161/450], Loss: 1.7273
Epoch [162/450], Loss: 1.7083
Epoch [163/450], Loss: 1.6803
Epoch [164/450], Loss: 1.7243
Epoch [165/450], Loss: 1.6958
Epoch [166/450], Loss: 1.5591
Epoch [167/450], Loss: 1.6995
Epoch [168/450], Loss: 1.7665
Epoch [169/450], Loss: 1.8992
Epoch [170/450], Loss: 1.8337
Epoch [171/450], Loss: 1.6950
Epoch [172/450], Loss: 1.7353
Epoch [173/450], Loss: 1.8955
Epoch [174/450], Loss: 1.7343
Epoch [175/450], Loss: 1.7595
Epoch [176/450], Loss: 1.7271
Epoch [177/450], Loss: 1.7138
Epoch [178/450], Loss: 1.6842
Epoch [179/450], Loss: 1.7432
Epoch [180/450], Loss: 1.6232
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Epoch [181/450], Loss: 1.6423
Epoch [182/450], Loss: 1.6902
Epoch [183/450], Loss: 1.5650
Epoch [184/450], Loss: 1.5680
Epoch [185/450], Loss: 1.7150
Epoch [186/450], Loss: 1.7247
Epoch [187/450], Loss: 1.9342
Epoch [188/450], Loss: 1.6885
Epoch [189/450], Loss: 1.6378
Epoch [190/450], Loss: 1.7386
Epoch [191/450], Loss: 1.7040
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Epoch [193/450], Loss: 1.8195
Epoch [194/450], Loss: 1.5564
Epoch [195/450], Loss: 1.7951
Epoch [196/450], Loss: 1.8352
Epoch [197/450], Loss: 1.7075
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Epoch [211/450], Loss: 1.5999
Epoch [212/450], Loss: 1.6851
Epoch [213/450], Loss: 1.6301
Epoch [214/450], Loss: 1.6067
Epoch [215/450], Loss: 1.5868
Epoch [216/450], Loss: 1.7703
Epoch [217/450], Loss: 1.5520
Epoch [218/450], Loss: 1.6060
Epoch [219/450], Loss: 1.7178
Epoch [220/450], Loss: 1.7587
Epoch [221/450], Loss: 1.6863
Epoch [222/450], Loss: 1.6025
Epoch [223/450], Loss: 2.0776
Epoch [224/450], Loss: 1.8999
Epoch [225/450], Loss: 1.9417
Epoch [226/450], Loss: 1.7662
Epoch [227/450], Loss: 1.8364
Epoch [228/450], Loss: 1.7876
Epoch [229/450], Loss: 1.6295
Epoch [230/450], Loss: 1.6889
Epoch [231/450], Loss: 1.8846
Epoch [232/450], Loss: 1.7550
Epoch [233/450], Loss: 1.6245
Epoch [234/450], Loss: 1.7895
Epoch [235/450], Loss: 1.7287
Epoch [236/450], Loss: 1.7353
Epoch [237/450], Loss: 1.7224
Epoch [238/450], Loss: 1.6644
Epoch [239/450], Loss: 1.8780
Epoch [240/450], Loss: 1.6519
```

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Epoch [241/450], Loss: 1.8437
Epoch [242/450], Loss: 1.7616
Epoch [243/450], Loss: 1.6209
Epoch [244/450], Loss: 1.6042
Epoch [245/450], Loss: 1.7965
Epoch [246/450], Loss: 1.7299
Epoch [247/450], Loss: 1.8922
Epoch [248/450], Loss: 1.6670
Epoch [249/450], Loss: 1.7096
Epoch [250/450], Loss: 1.6400
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Epoch [252/450], Loss: 1.8658
Epoch [253/450], Loss: 2.0656
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Epoch [255/450], Loss: 1.6021
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Epoch [257/450], Loss: 1.6347
Epoch [258/450], Loss: 1.6894
Epoch [259/450], Loss: 1.8338
Epoch [260/450], Loss: 1.8199
Epoch [261/450], Loss: 1.7293
Epoch [262/450], Loss: 1.6780
Epoch [263/450], Loss: 1.8552
Epoch [264/450], Loss: 1.7938
Epoch [265/450], Loss: 1.7079
Epoch [266/450], Loss: 1.7427
Epoch [267/450], Loss: 1.7079
Epoch [268/450], Loss: 1.7837
Epoch [269/450], Loss: 1.6572
Epoch [270/450], Loss: 1.6993
Epoch [271/450], Loss: 1.6801
Epoch [272/450], Loss: 1.9474
Epoch [273/450], Loss: 2.2106
Epoch [274/450], Loss: 1.9627
Epoch [275/450], Loss: 1.6634
Epoch [276/450], Loss: 1.6487
Epoch [277/450], Loss: 1.7150
Epoch [278/450], Loss: 1.8457
Epoch [279/450], Loss: 1.6798
Epoch [280/450], Loss: 1.7400
Epoch [281/450], Loss: 1.6900
Epoch [282/450], Loss: 1.6830
Epoch [283/450], Loss: 1.8201
Epoch [284/450], Loss: 1.6626
Epoch [285/450], Loss: 1.8965
Epoch [286/450], Loss: 1.7760
Epoch [287/450], Loss: 1.8362
Epoch [288/450], Loss: 1.6490
Epoch [289/450], Loss: 1.9476
Epoch [290/450], Loss: 1.8473
Epoch [291/450], Loss: 1.7434
Epoch [292/450], Loss: 1.5816
Epoch [293/450], Loss: 1.8366
Epoch [294/450], Loss: 1.9580
Epoch [295/450], Loss: 1.9717
Epoch [296/450], Loss: 1.7484
Epoch [297/450], Loss: 1.6250
Epoch [298/450], Loss: 1.5829
Epoch [299/450], Loss: 1.6767
Epoch [300/450], Loss: 1.6164
```

Epoch [301/450], Loss: 1.9012
Epoch [302/450], Loss: 1.8056
Epoch [303/450], Loss: 1.6738
Epoch [304/450], Loss: 1.5883
Epoch [305/450], Loss: 1.7126
Epoch [306/450], Loss: 1.6266
Epoch [307/450], Loss: 1.6167
Epoch [308/450], Loss: 1.8619
Epoch [309/450], Loss: 1.6779
Epoch [310/450], Loss: 1.7510
Epoch [311/450], Loss: 1.6698
Epoch [312/450], Loss: 1.6961
Epoch [313/450], Loss: 1.6315
Epoch [314/450], Loss: 1.6069
Epoch [315/450], Loss: 1.6191
Epoch [316/450], Loss: 1.5216
Epoch [317/450], Loss: 1.6490
Epoch [318/450], Loss: 1.6895
Epoch [319/450], Loss: 1.9167
Epoch [320/450], Loss: 1.7245
Epoch [321/450], Loss: 1.7396
Epoch [322/450], Loss: 1.9328
Epoch [323/450], Loss: 1.8863
Epoch [324/450], Loss: 1.6003
Epoch [325/450], Loss: 1.6263
Epoch [326/450], Loss: 1.7353
Epoch [327/450], Loss: 2.0424
Epoch [328/450], Loss: 1.9368
Epoch [329/450], Loss: 1.8860
Epoch [330/450], Loss: 2.1553
Epoch [331/450], Loss: 1.8803
Epoch [332/450], Loss: 1.7709
Epoch [333/450], Loss: 1.6843
Epoch [334/450], Loss: 1.6773
Epoch [335/450], Loss: 1.8707
Epoch [336/450], Loss: 1.9643
Epoch [337/450], Loss: 1.6869
Epoch [338/450], Loss: 1.5245
Epoch [339/450], Loss: 1.6704
Epoch [340/450], Loss: 1.7036
Epoch [341/450], Loss: 1.6285
Epoch [342/450], Loss: 1.6808
Epoch [343/450], Loss: 1.6705
Epoch [344/450], Loss: 1.7661
Epoch [345/450], Loss: 1.7865
Epoch [346/450], Loss: 1.7642
Epoch [347/450], Loss: 1.5878
Epoch [348/450], Loss: 1.5221
Epoch [349/450], Loss: 1.5938
Epoch [350/450], Loss: 1.7261
Epoch [351/450], Loss: 1.5151
Epoch [352/450], Loss: 1.5449
Epoch [353/450], Loss: 1.5439
Epoch [354/450], Loss: 1.5637
Epoch [355/450], Loss: 1.6013
Epoch [356/450], Loss: 1.5845
Epoch [357/450], Loss: 1.7391
Epoch [358/450], Loss: 1.5649
Epoch [359/450], Loss: 1.6631
Epoch [360/450], Loss: 1.8304

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Epoch [361/450], Loss: 1.7086
Epoch [362/450], Loss: 1.4894
Epoch [363/450], Loss: 1.7533
Epoch [364/450], Loss: 1.9968
Epoch [365/450], Loss: 1.6922
Epoch [366/450], Loss: 1.5835
Epoch [367/450], Loss: 1.5205
Epoch [368/450], Loss: 1.8384
Epoch [369/450], Loss: 1.7516
Epoch [370/450], Loss: 1.7342
Epoch [371/450], Loss: 1.7708
Epoch [372/450], Loss: 1.7679
Epoch [373/450], Loss: 1.6571
Epoch [374/450], Loss: 1.6114
Epoch [375/450], Loss: 1.5672
Epoch [376/450], Loss: 1.5844
Epoch [377/450], Loss: 1.7258
Epoch [378/450], Loss: 1.7720
Epoch [379/450], Loss: 1.8129
Epoch [380/450], Loss: 1.8325
Epoch [381/450], Loss: 1.6550
Epoch [382/450], Loss: 1.7127
Epoch [383/450], Loss: 1.6777
Epoch [384/450], Loss: 1.8628
Epoch [385/450], Loss: 1.6138
Epoch [386/450], Loss: 1.6764
Epoch [387/450], Loss: 1.6915
Epoch [388/450], Loss: 1.6663
Epoch [389/450], Loss: 1.7935
Epoch [390/450], Loss: 1.6561
Epoch [391/450], Loss: 1.5402
Epoch [392/450], Loss: 1.6958
Epoch [393/450], Loss: 1.6358
Epoch [394/450], Loss: 1.7893
Epoch [395/450], Loss: 1.6398
Epoch [396/450], Loss: 1.5882
Epoch [397/450], Loss: 1.6703
Epoch [398/450], Loss: 1.7428
Epoch [399/450], Loss: 1.9232
Epoch [400/450], Loss: 2.0015
Epoch [401/450], Loss: 1.7575
Epoch [402/450], Loss: 1.6398
Epoch [403/450], Loss: 1.6880
Epoch [404/450], Loss: 1.7546
Epoch [405/450], Loss: 1.8921
Epoch [406/450], Loss: 1.7405
Epoch [407/450], Loss: 1.5991
Epoch [408/450], Loss: 1.6139
Epoch [409/450], Loss: 1.6092
Epoch [410/450], Loss: 1.5520
Epoch [411/450], Loss: 1.6004
Epoch [412/450], Loss: 1.7077
Epoch [413/450], Loss: 2.0108
Epoch [414/450], Loss: 1.7082
Epoch [415/450], Loss: 1.6331
Epoch [416/450], Loss: 1.7132
Epoch [417/450], Loss: 1.6089
Epoch [418/450], Loss: 1.7009
Epoch [419/450], Loss: 1.9630
Epoch [420/450], Loss: 1.8758
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Epoch [421/450], Loss: 1.8016
Epoch [422/450], Loss: 1.6454
Epoch [423/450], Loss: 1.6091
Epoch [424/450], Loss: 1.8184
Epoch [425/450], Loss: 1.6220
Epoch [426/450], Loss: 1.7547
Epoch [427/450], Loss: 1.5815
Epoch [428/450], Loss: 1.6116
Epoch [429/450], Loss: 1.5475
Epoch [430/450], Loss: 1.7481
Epoch [431/450], Loss: 1.5297
Epoch [432/450], Loss: 1.4840
Epoch [433/450], Loss: 1.6852
Epoch [434/450], Loss: 1.6521
Epoch [435/450], Loss: 1.6281
Epoch [436/450], Loss: 1.7643
Epoch [437/450], Loss: 1.6712
Epoch [438/450], Loss: 1.7913
Epoch [439/450], Loss: 1.7048
Epoch [440/450], Loss: 1.7146
Epoch [441/450], Loss: 1.7021
Epoch [442/450], Loss: 1.7280
Epoch [443/450], Loss: 1.9074
Epoch [444/450], Loss: 1.6420
Epoch [445/450], Loss: 1.5342
Epoch [446/450], Loss: 1.6065
Epoch [447/450], Loss: 1.6330
Epoch [448/450], Loss: 1.7487
Epoch [449/450], Loss: 1.6621
Epoch [450/450], Loss: 1.6064
Multi-layer model test loss: 3.6781

```

Problem 2

Derive a hypothesis class H and a data distribution P such that there exists a hypothesis $h \in H$ for which the training error is zero, but the expected test error is no better than random guessing.

1. Define a finite hypothesis class H with at least two hypotheses and describe a data distribution P on (X, Y) .
2. Construct a small training set drawn from P where one hypothesis overfit fits the training data perfectly, achieving zero training error.
3. Explain why, under the chosen P , this hypothesis does no better than random guessing on unseen test data.

Solution

Let P be a distribution on (X, Y) ,
 where $X \in \mathbb{R}^2$ and $Y \in \{0, 1\}$ is a binary label.

$X = (X_1, X_2)$ there exists a boundary in $X_{1,boundary}$ and $X_{2,boundary}$ such that:

- If $X_1 \geq X_{1,boundary}$ and $X_2 \geq X_{2,boundary}$, then $Y = 0$.
- If $X_1 \leq X_{1,boundary}$ and $X_2 \leq X_{2,boundary}$, then $Y = 0$.

- For all other combinations, $Y = 1$.

Then,

1. Hypothesis class H may contain:

- $h_1(X) = 0$ if $X_1 \geq X_{1,boundary}$ else 1
- $h_2(X) = 0$ if $X_1 \leq X_{1,boundary}$

2. Let the training set S_N be defined as:

$$S_{N,smallregion} = \{((X_1, X_2), Y) | X_2 < X_{2,boundary}\}$$

then, h_2 fits perfectly.

3. However, if a better training set is constructed as:

$$S_{N,largeregion} = \{((X_1, X_2), Y) | X_{1,boundary} - \delta < X_1 < X_{1,boundary} + \delta, X_{2,boundary} < X_2 < X_{2,boundary} + \delta\}$$

h_2 or h_1 will perform no better than random.

Problem 3:

Need to show that

$$R(\hat{h}_n) \leq R(h^*) + 2\epsilon$$

or equivalently,

$$R(\hat{h}_n) - R(h^*) \leq 2\epsilon$$

Start with LHS, and add and subtract $R(\hat{h}_n)$ and $\hat{R}(h^*)$:

$$R(\hat{h}_n) - R(h^*) = [R(\hat{h}_n) - \hat{R}_n(\hat{h}_n)] + [\hat{R}_n(\hat{h}_n) - \hat{R}_n(h^*)] + [\hat{R}_n(h^*) - R(h^*)]$$

On the RHS, for term 1, we can use Uniform Convergence for Finite Hypothesis Class Theorem for $h = \hat{h}_n$. And get $R(\hat{h}_n) - \hat{R}_n(\hat{h}_n) \leq \epsilon$.

and term 2, we can use the fact that $\hat{h}_n$ gives the minimum of $\hat{R}_n(h)$ than any other h , so $\hat{R}_n(\hat{h}_n) \leq \hat{R}_n(h^*)$ or $\hat{R}_n(\hat{h}_n) - \hat{R}_n(h^*) \leq 0$.

and for term 3, again applying Uniform Convergence for Finite Hypothesis Class Theorem for $h = h^*$, we get $\hat{R}_n(h^*) - R(h^*) \leq \epsilon$.

Finally,

$$R(\hat{h}_n) - R(h^*) \leq \epsilon + 0 + \epsilon$$

$$R(\hat{h}_n) - R(h^*) \leq 2\epsilon$$